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THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 4: Modeling based on First-principle

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Modeling Methods

- **Mathematical Model**

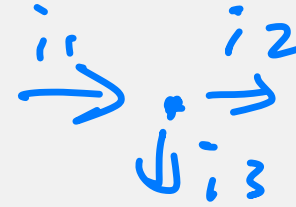
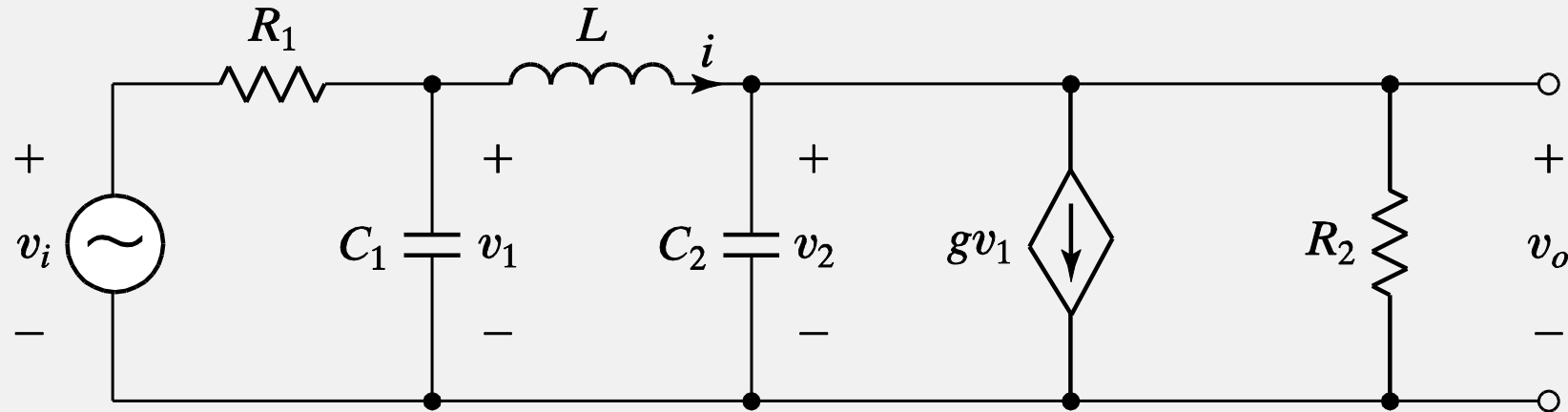
The **relationships** among various physical variables in a system, in particular that between the **input** and the **output** of the system.

- **Modeling Methods**

- **First principle modeling:** writing down differential equations based on physical laws.
- **System identification:** finding out the relationship between input and output from a set of experimental data.
- **Combine** the above two methods

Electrical Systems

- A simple RLC circuit



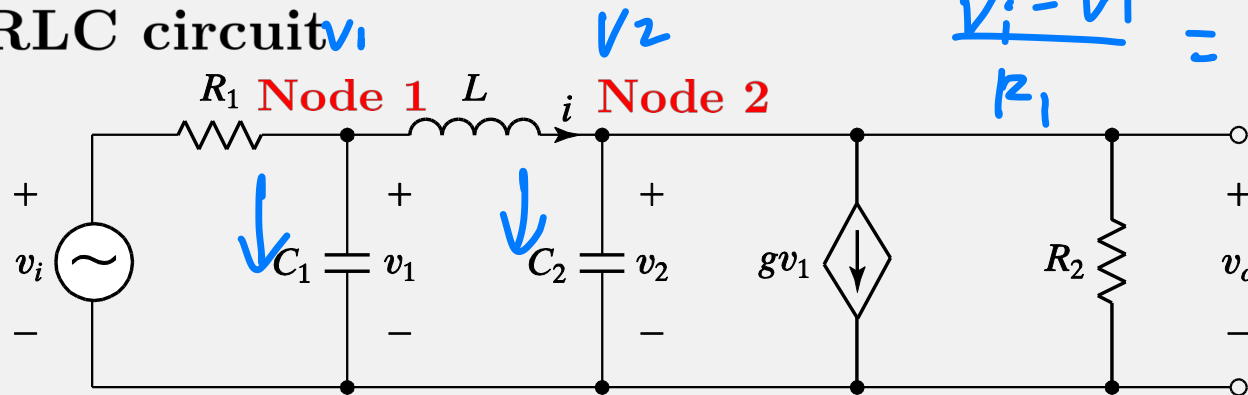
$$\sum i = 0$$
$$i_1 = i_2 + i_3$$

$$\sum v = 0$$

- The **input** $v_i(t)$ and **output** $v_o(t)$
- **Intermediate variables:** capacitor voltages $v_1(t)$ and $v_2(t)$,
inductor current $i(t)$
- **Physical laws:** Kirchhoff's current law (KCL),
Kirchhoff's voltage law (KVL)

Electrical Systems

- A simple RLC circuit



- Apply KCL to **Node 1** connecting R_1 , C_1 , and L :

$$C_1 \frac{dv_1(t)}{dt} = \frac{v_i(t) - v_1(t)}{R_1} - i(t).$$

- Apply KCL to **Node 2** connecting L , C_2 , R_2 , and the controlled source:

$$C_2 \frac{dv_2(t)}{dt} = i(t) - g v_1(t) - \frac{v_2(t)}{R_2}.$$

$$\dot{v}_1 = \frac{v_i}{R_1 C_1} - \frac{v_1}{R_1 C_1} - \frac{i}{C_1}$$

$$v_1 - v_2 = L \frac{di}{dt}$$

$$\frac{v_i - v_1}{R_1} = i + C_2 \frac{dv_2}{dt}$$

$$i = C_2 \frac{dv_2}{dt} + g v_1 + \frac{v_2}{R_2}$$

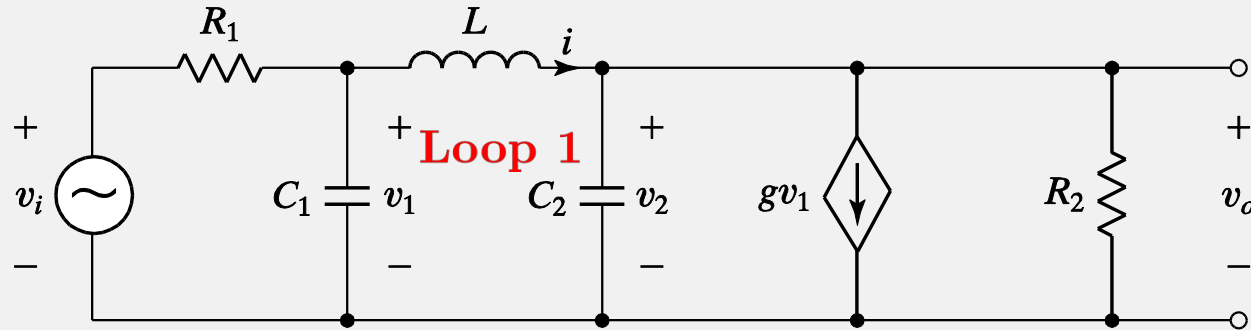
$$v_2 = i / \left(\frac{1}{C_2} - g v_1 / C_2 - v_2 / (R_2 C_2) \right)$$

$$\begin{pmatrix} \dot{v}_1 \\ \dot{v}_2 \\ i \end{pmatrix} = \begin{pmatrix} 1/R_1 C_1 & 0 & -1/C_1 \\ -g/C_2 & -1/R_2 C_2 & 1/C_2 \\ 1/L & -1/L & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ i \end{pmatrix}$$

$$v_o = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ i \end{pmatrix} + \begin{pmatrix} 1/R_1 C_1 \\ 0 \\ 0 \end{pmatrix} v_i$$

Electrical Systems

- A simple RLC circuit



- Apply KVL to **Loop 1** containing C_1 , L , and C_2 :

$$L \frac{di(t)}{dt} = v_1(t) - v_2(t).$$

- The output $v_o(t)$ is simply one of the intermediate variables:

$$v_o(t) = v_2(t).$$

Electrical Systems

- A simple RLC circuit

The **differential equations** as well as the **output** give a model of the RLC circuit. The model can also be organized in a **matrix form**

$$\begin{bmatrix} C_1 & 0 & 0 \\ 0 & C_2 & 0 \\ 0 & 0 & L \end{bmatrix} \begin{bmatrix} \dot{v}_1(t) \\ \dot{v}_2(t) \\ \dot{i}(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1} & 0 & -1 \\ -g & -\frac{1}{R_2} & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \\ i(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1} \\ 0 \\ 0 \end{bmatrix} v_i(t),$$
$$v_o(t) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \\ i(t) \end{bmatrix}.$$

Electrical Systems

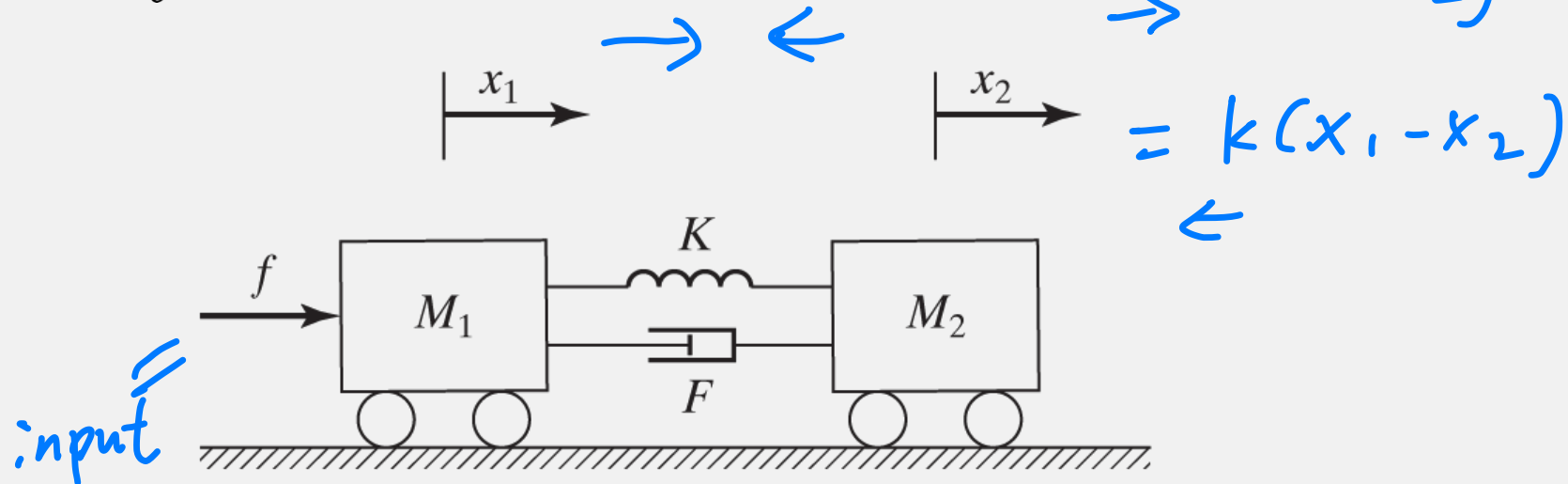
- A simple RLC circuit

Premultiply both sides by the inverse of the matrix on the left-hand side the equation. We then obtain a **state space model**:

$$\begin{bmatrix} \dot{v}_1(t) \\ \dot{v}_2(t) \\ \dot{i}(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C_1} & 0 & -\frac{1}{C_1} \\ -\frac{g}{C_2} & -\frac{1}{R_2 C_2} & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \\ i(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1 C_1} \\ 0 \\ 0 \end{bmatrix} v_i(t),$$
$$v_o(t) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \\ i(t) \end{bmatrix}.$$

Mechanical Systems

- A two-cart system



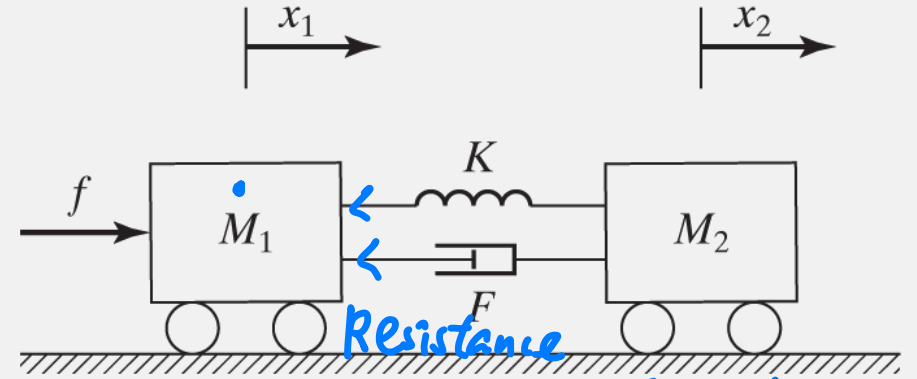
- The **input** $f(t)$ and **output** $y(t) = x_2(t)$

$$M\ddot{x}_1 = f - k(x_1 - x_2) - F$$

- **Intermediate variables:** the positions of the carts $x_1(t)$, $x_2(t)$
- **Physical laws:** Newton's second law

Mechanical Systems

- A two-cart system
- Apply Newton's second law to the **two carts**:



$$M_1 \frac{d^2 x_1(t)}{dt^2} = \boxed{f(t)} - K(x_1(t) - x_2(t)) - F \left(\frac{dx_1(t)}{dt} - \frac{dx_2(t)}{dt} \right),$$

$$M_2 \frac{d^2 x_2(t)}{dt^2} = K(x_1(t) - x_2(t)) + F \left(\frac{dx_1(t)}{dt} - \frac{dx_2(t)}{dt} \right).$$

$$F_1 = kx$$

$$F_2 = F_1 v$$

$$M_1 \ddot{x}_1 = f - k(x_1 - x_2)$$

- Reorganize the equations into a **matrix form**

$$M_2 \ddot{x}_2 = k(x_1 - x_2) + F(\dot{x}_1 - \dot{x}_2)$$

$$\begin{bmatrix} M_1 & 0 \\ 0 & M_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1(t) \\ \ddot{x}_2(t) \end{bmatrix} + \begin{bmatrix} F & -F \\ -F & F \end{bmatrix} \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} + \begin{bmatrix} K & -K \\ -K & K \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} f(t).$$

output = x_2

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/M_1 & k/M_1 - F/M_1 & F/M_1 & 0 \\ k/M_2 & -k/M_2 & F/M_2 & -F/M_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1/M_1 \\ 0 \end{pmatrix} f$$

$$y = (0 \ 1 \ 0 \ 0) \begin{pmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{pmatrix}$$

Mechanical Systems

- A two-cart system

Define the state vector $\mathbf{x}$, and input vector $\mathbf{u}$:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \dot{x}_1 \\ x_2 \\ \dot{x}_2 \end{bmatrix}, \quad \mathbf{u} = [f]$$

The state space model:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

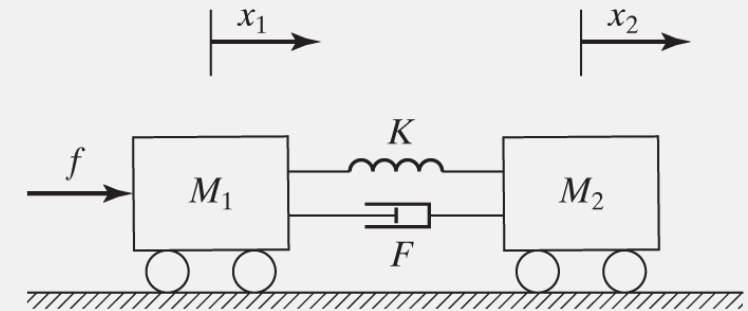
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k}{M_1} & -\frac{F}{M_1} & \frac{k}{M_1} & \frac{F}{M_1} \\ 0 & 0 & 0 & 1 \\ \frac{k}{M_2} & \frac{F}{M_2} & -\frac{k}{M_2} & -\frac{F}{M_2} \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{M_1} \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

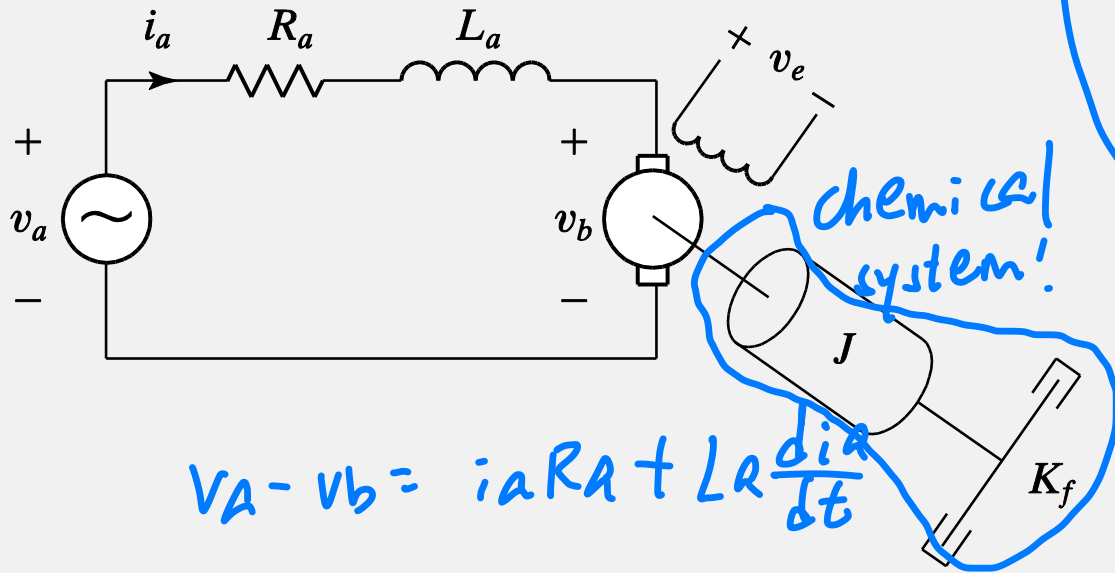
$$\mathbf{D} = [0]$$



Design output from myself

Electromechanical Systems

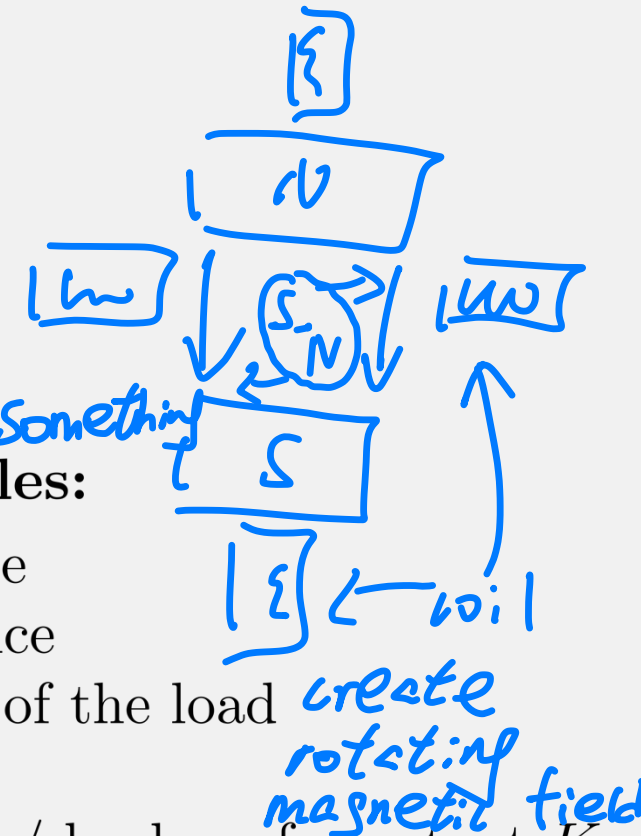
- An armature-controlled DC motor system



Resistance for rotating something

★Parameters and variables:

R_a :	armature resistance
L_a :	armature inductance
J :	moment of inertia of the load
K_f :	friction coefficient
K :	torque constant K_t / back emf constant K_b
$v_a(t)$:	armature voltage
$i_a(t)$:	armature current
$v_b(t)$:	back electro-motive force (back emf)
$\tau_m(t)$:	motor torque
$\theta(t)$:	angular position of the motor shaft
$\omega(t)$:	angular velocity of the motor shaft ($= \dot{\theta}(t)$)





$$F = ma$$

$$p = mv$$



$$\tau = J \cdot \alpha$$

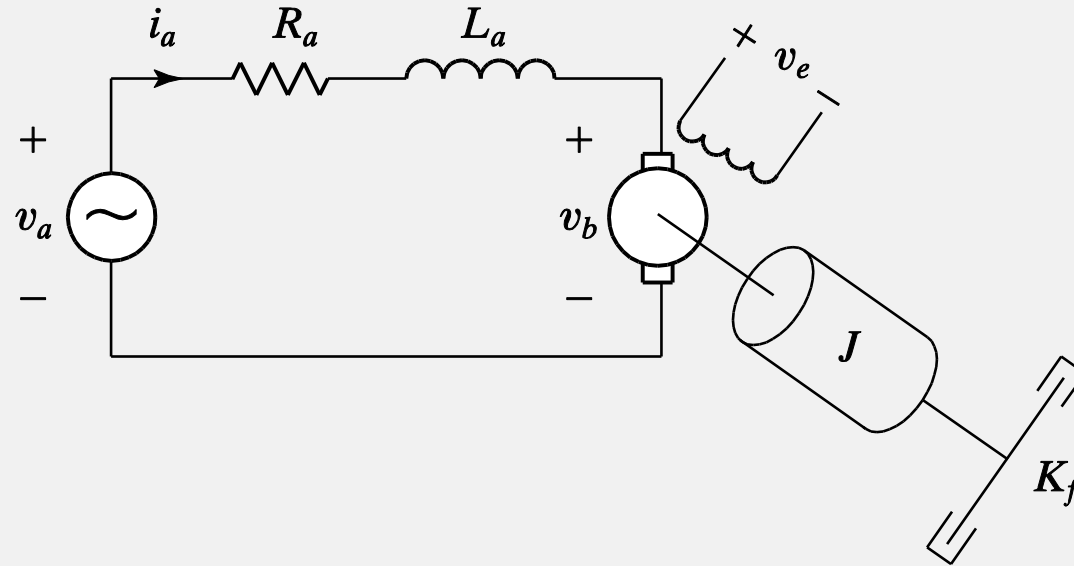
$$L = J \omega$$

Angular momentum

$$J = \int r^2 dm$$

Electromechanical Systems

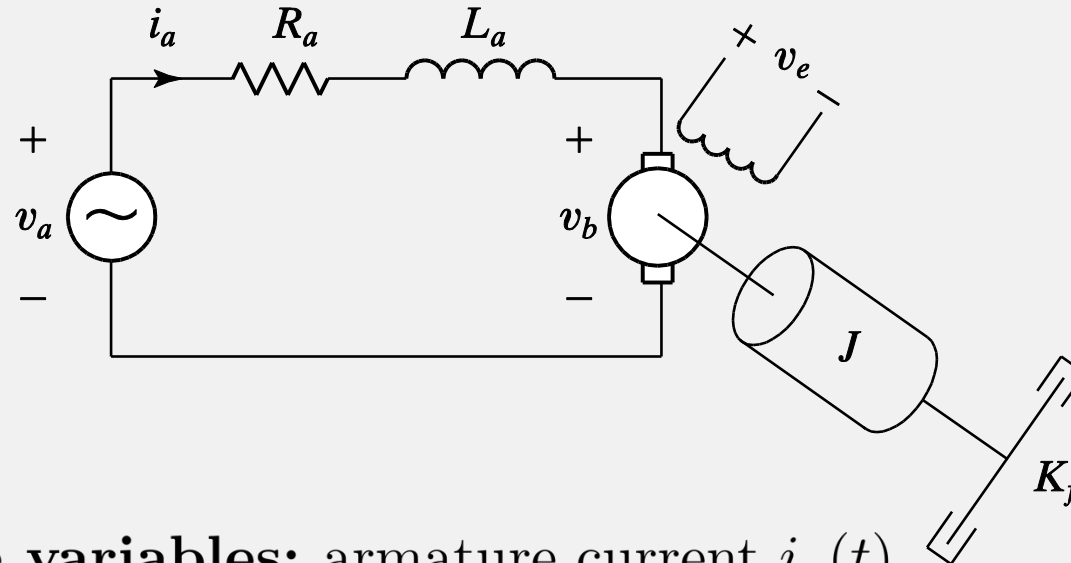
- An armature-controlled DC motor system



- The **input** armature voltage $v_a(t)$
- The **output**
 - **position control** case: angular position of the motor shaft $\theta(t)$
 - **speed control** case: angular velocity of the motor shaft $\omega(t)$ *speed control*

Electromechanical Systems

- An armature-controlled DC motor system



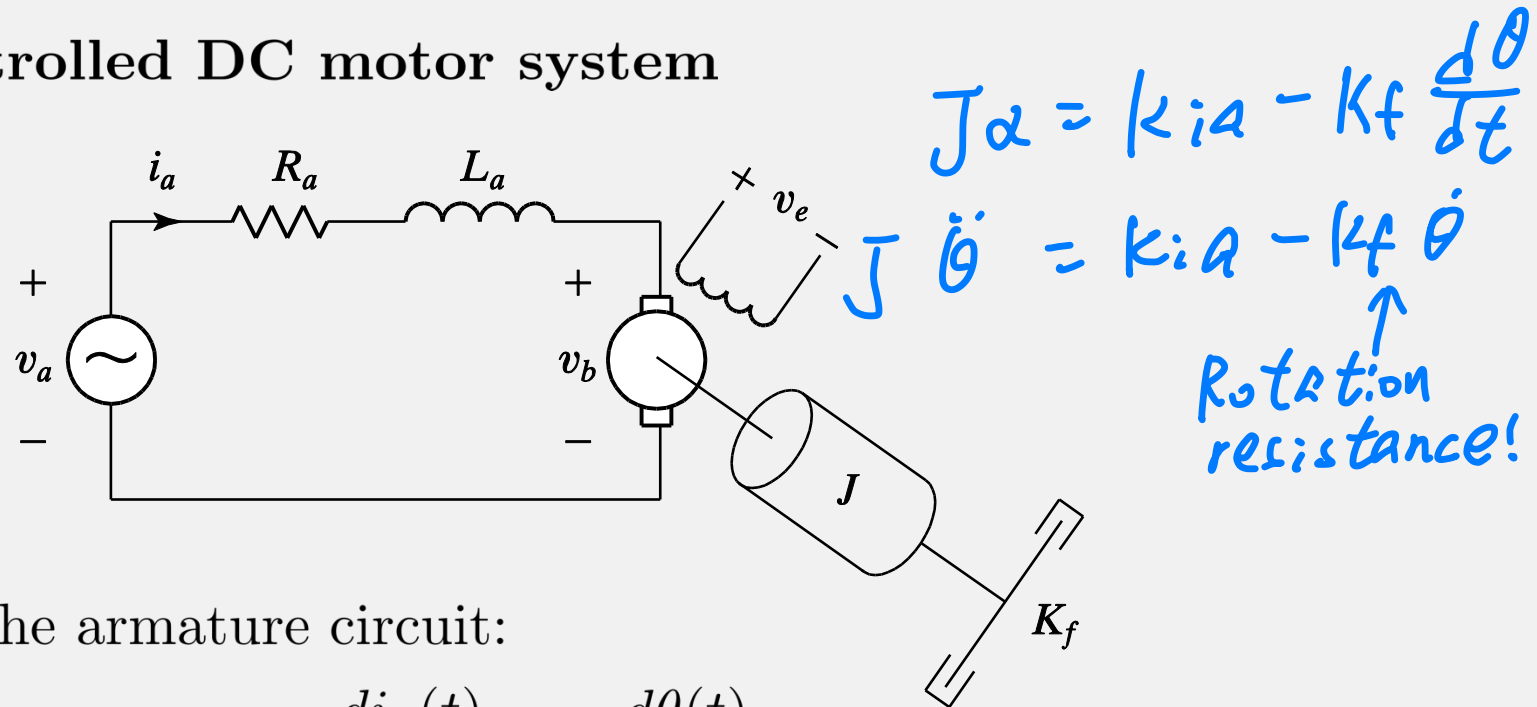
$$v_a = i_a R_a + L_a \frac{di_a}{dt} + K \frac{d\theta}{dt}$$

↑
angular velocity

- **Intermediate variables:** armature current $i_a(t)$,
angular position $\theta(t)$ or angular velocity $\omega(t)$
- **Physical laws:** Kirchhoff's voltage law (KVL),
the rotational version of Newton's second law

Electromechanical Systems

- An armature-controlled DC motor system



- KVL applied to the armature circuit:

$$R_a i_a(t) + L_a \frac{di_a(t)}{dt} + K \frac{d\theta(t)}{dt} = v_a(t).$$

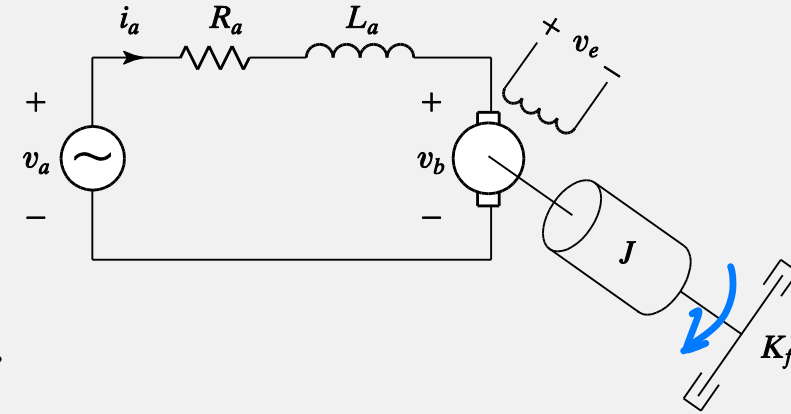
- The rotational version of Newton's second law applied to the motor shaft:

$$J \frac{d^2\theta(t)}{dt^2} = K i_a(t) - K_f \frac{d\theta(t)}{dt}.$$

Position control

Electromechanical Systems

- **System model:** $R_a i_a(t) + L_a \frac{di_a(t)}{dt} + K \frac{d\theta(t)}{dt} = v_a(t),$
 $K i_a(t) - K_f \frac{d\theta(t)}{dt} = J \frac{d^2\theta(t)}{dt^2}.$



- Choose the state variables, input and output variables as

$$x_1(t) = i_a(t), x_2(t) = \theta(t), x_3(t) = \omega(t), u(t) = v_a(t), y(t) = \theta(t).$$

- **State space model:**

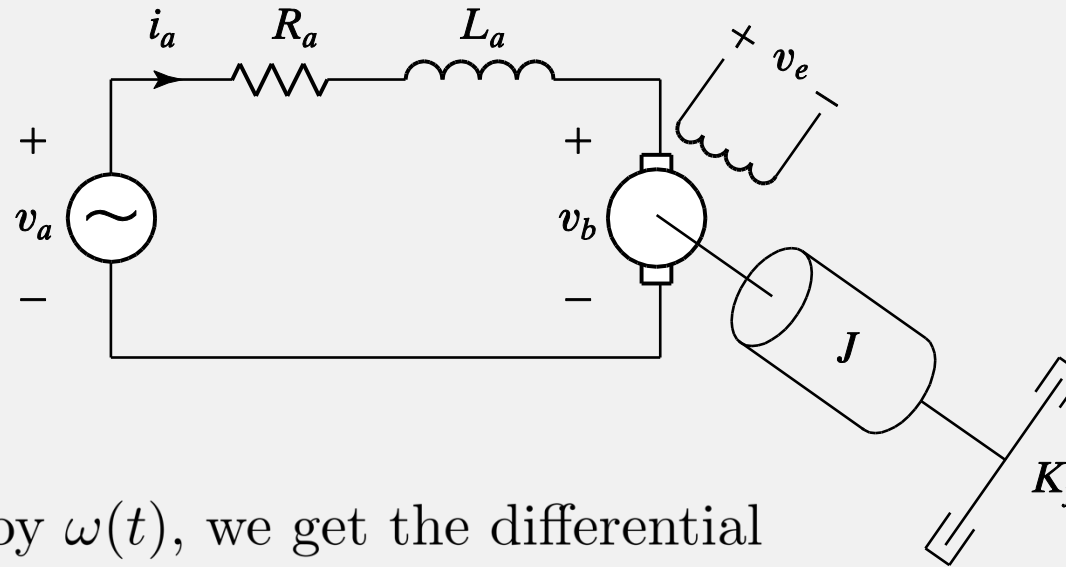
$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -R_a/L_a & 0 & -K/L_a \\ 0 & 0 & 1 \\ K/J & 0 & -K_f/J \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1/L_a \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$\begin{pmatrix} i_a \\ \theta \\ \dot{\theta} \end{pmatrix}$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}.$$

Electromechanical Systems

- An armature-controlled DC motor system



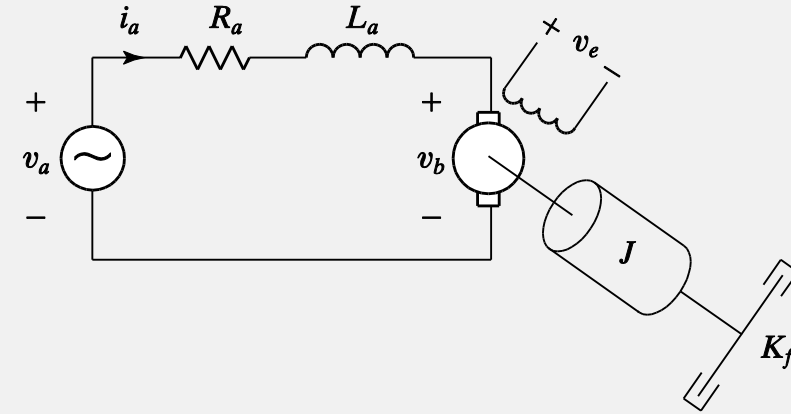
- Replacing $\frac{d\theta(t)}{dt}$ by $\omega(t)$, we get the differential equation model of a DC motor system in the **speed control** case.

$$R_a i_a(t) + L_a \frac{di_a(t)}{dt} + K\omega(t) = v_a(t),$$

$$J \frac{d\omega(t)}{dt} = K i_a(t) - K_f \omega(t).$$

Electromechanical Systems

- **System model:** $R_a i_a(t) + L_a \frac{di_a(t)}{dt} + K\omega(t) = v_a(t),$
 $K i_a(t) - K_f \omega(t) = J \frac{d\omega(t)}{dt}.$



$(\dot{\omega})$

- Choose the state variables, input and output variables as

$$x_1(t) = i_a(t), x_2(t) = \omega(t), u(t) = v_a(t), y(t) = \omega(t).$$

- **State space model:**

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -K/L_a \\ K/J & -K_f/J \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1/L_a \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

Review & Exercise

- How to determine the input and output variables?
- How to apply KCL, KVL and Newton's Law to establish a model for a system?
- How to convert differential equations into Matrix?
- As you can see, the system modeling requires some physical and mathematic skills. Please review the relevant parts and do some exercises by yourself if you are not familiar with it.

Appendix

Electromechanical Systems

K : torque constant K_t / back emf constant K_b

When the motor shaft turns, the magnetic field generates a potential in the rotor winding as a result of the Faraday induction. This potential is called the back electro-motive force (back emf). There are **two basic relations** in a DC motor:

One is that the torque in the motor shaft is proportional to the armature current via the torque constant K_t , i.e.,

$$\tau_m(t) = K_t i_a(t)$$

The other is that the back emf $v_b(t)$ is proportional to the motor velocity $\omega(t)$ via the back emf constant K_b , i.e.,

$$v_b(t) = K_b \omega(t)$$

The electrical power consumed by the DC motor for the electrical to mechanical energy conversion is $i_a(t)v_b(t)$ in watt [W] and the mechanical power of the DC motor is $\tau_m(t)\omega(t)$ also in watt. Due to the conservation of energy, we must have

$$\begin{aligned} i_a(t)v_b(t) &= \tau_m(t)\omega(t), \\ \Downarrow \\ i_a(t)K_b\omega(t) &= K_t i_a(t)\omega(t). \end{aligned}$$

This shows that $K_t = K_b$ with units Nm/A and V sec/Rad, which are the same. Therefore, we can use constant K to denote both K_t and K_b .

